Vegetation of Peninsular Malaysia.

Chapter · August 2010

DOI: 10.13140/RG.2.1.2072.8486

Vegetation of Peninsular Malaysia

L.G. Saw

Forest Research Institute Malaysia,
Kepong, Malaysia

Introduction

Peninsular Malaysia is at the southern extremity of the Asian continent and covers an area of 131,598 km^2. It separates the South China Sea from the Straits of Malacca. The region is generally hilly or mountainous and over 40% of the land is above 150 m a.s.l. with 23% over 300 m (Wyatt-Smith, 1963). The mountains run in a series of ranges in a north-south direction (see map on inside front cover). The largest of these, the Main (Titiwangsa) Range, is a continuous granitic range extending from beyond the Perak-Thailand border to the Negeri Sembilan-Melaka boundary near Tampin. The highest peak of the Main Range is G. Korbu (2,182 m). Other important ranges include the Perlis limestone ridge along the west Perlis-Thailand border; the west Kedah and Province Wellesley range; the Bintang range; the Keladang Saiong range running about 160 km parallel to Sg. Perak and between the Bintang and the Main Range; the Benom range of west-central Pahang rises to its highest peak on G. Benom (2,130 m) extending south to G. Ledang (1,276 m) in Johor; the Tahan range in central Pahang with G. Tahan (2,188 m), the highest mountain in Peninsular Malaysia, culminates on the Pahang-Kelantan border in the north and extends into central Johor to the south; and finally a dissected long granitic block in Terengganu forms the boundary with Kelantan and Pahang. The four main rivers are Sg. Pahang (500 km) and Sg. Kelantan (250 km) draining into the South China Sea (Tjia, 1988), and Sg. Perak (390 km) and Sg. Muar (190 km) draining into the Straits of Malacca.

Forest cover in Peninsular Malaysia is still relatively extensive in spite of widespread land conversion starting in the early 20th century with coffee plantations and tin mining followed by the large scale plantations of rubber and, in the later part of the century, oil palm. Forest cover now stands at 40.7% of the total land area (Table 1). This figure has now more-or-less stabilised. Within these forested areas, the National/State Parks and the Wildlife and Bird Sanctuaries are the best protected areas with no logging allowed. For the Permanent Reserved Forests (PFR), there are two categories: the production forest that is subjected to forestry management for timber production (this constitutes 21.3% of the land area), and the protection forest where, once an area is listed, it will not be subject to logging. The protection forests consist of both primary forest areas, such as Virgin Jungle Reserves (VJR), and areas that have been logged but subsequently identified for conservation or for environmental protection purposes. These include High Value Conservation Forest (HVCF), mainly for the protection of plant and animal species, areas with steep slopes, mountain areas above 1,000 m and amenity forests (for recreation). Such areas cover about 14.4% of the total area in Peninsular Malaysia.

Table 1. Forest cover and protected area systems according to different land use categories for Peninsular Malaysia (Anon., 2008).

Categories		Area (million ha)	Percentage
Permanent Reserved Forests (PFR)	Protection forest	1.90	14.4
	Production forest	2.80	21.3
National/State Parks		0.58[+]	4.4
Wildlife and Bird Sanctuaries		0.31[*]	2.4
Total Forest Cover		5.36	40.7
Other land use (agriculture and built-up areas)		7.80	59.3
Total Land Area		13.16	100

[+] A total of 0.04 million ha is located within PFR
[*] A total of 0.19 million ha is located within PFR

Factors Affecting Vegetation

Climate, soil, and soil water are the main interacting environmental factors that influence the distribution of plants and vegetation. Peninsular Malaysia located within a narrow latitudinal range of between 6°43’ N and 1°15’ N has a climate that is rather uniform throughout the year. It is generally regarded as perhumid (i.e., wet throughout the year) with short dry periods. The only exception is in north-west Peninsular Malaysia that experiences a few dry months each year. Two annual alternating monsoons determine the rainfall patterns: the north-east and south-west monsoons. The annual north-east monsoon brings a greater amount of rain in comparison to the south-west. The annual mean rainfall in Peninsular Malaysia is approximately 2,540 mm. In the wettest parts, annual rainfall exceeds 2,750 mm (Tjia, 1988), with the heaviest rainfall occurring on the foothills of Terengganu (average 4,000 mm per year).

The mean annual temperatures in the lowlands vary within 1.6°C of 26.7°C. Elsewhere ambient temperature limits the distribution of many plants and animals but in Malaysia, the only major influence on temperature is changes in temperature resulting from change in altitude, with a temperature drop of 6.5°C with every 1 km increase in altitude. For example, the daily temperatures at Cameron Highlands, Pahang (at 1,450 m a.s.l.) average a minimum range of 13–14°C and a maximum range of 22–23°C. This is on the average at least 10°C cooler than the lowlands. At these lower average temperatures, the vegetation is very different from that of the lowlands. Most rainforest formations developed over altitudinal changes (with the exception of alpine vegetation) are found in Peninsular Malaysia (Table 2).

Soils are another key determining factor in vegetation development. Soils in Peninsular Malaysia are generally acidic, predominantly weathered from igneous rocks (granite) into oxisols and ultisols. The alluvial soils that accumulate in river valley systems are also mainly soils washed down from hills. Soils in Peninsular Malaysia are generally deep compared to those in Sabah and Sarawak (Ashton, 1995). The major forest formations are all found on these soils. Podzols occur in places where parent materials are predominantly of quartz. These materials are low in clay and bases and also in minerals that might weather into these products. Such soils that develop from sandstone bedrocks or in raised sandy beaches running parallel to the sea coast (permatangs) are rather common on the East Coast of Peninsular Malaysia. Limestone formations in Peninsular Malaysia are not extensive. Most limestone hills are tower karsts, rising abruptly and most reaching about 500 m with Gua Peningat, Pahang, reaching 713 m. Other soil parent material or rock formations that can influence plant distribution are quartzite and ultramafic rocks. Quartzite is not extensive in Peninsular Malaysia and the small area of ultramafic rock in Peninsular Malaysia has no influence on vegetation because it is overlain by alluvial soils.

Table 2. Forest formations of tropical moist forests (after Whitmore, 1984) with modifications for Peninsular Malaysian vegetation types.

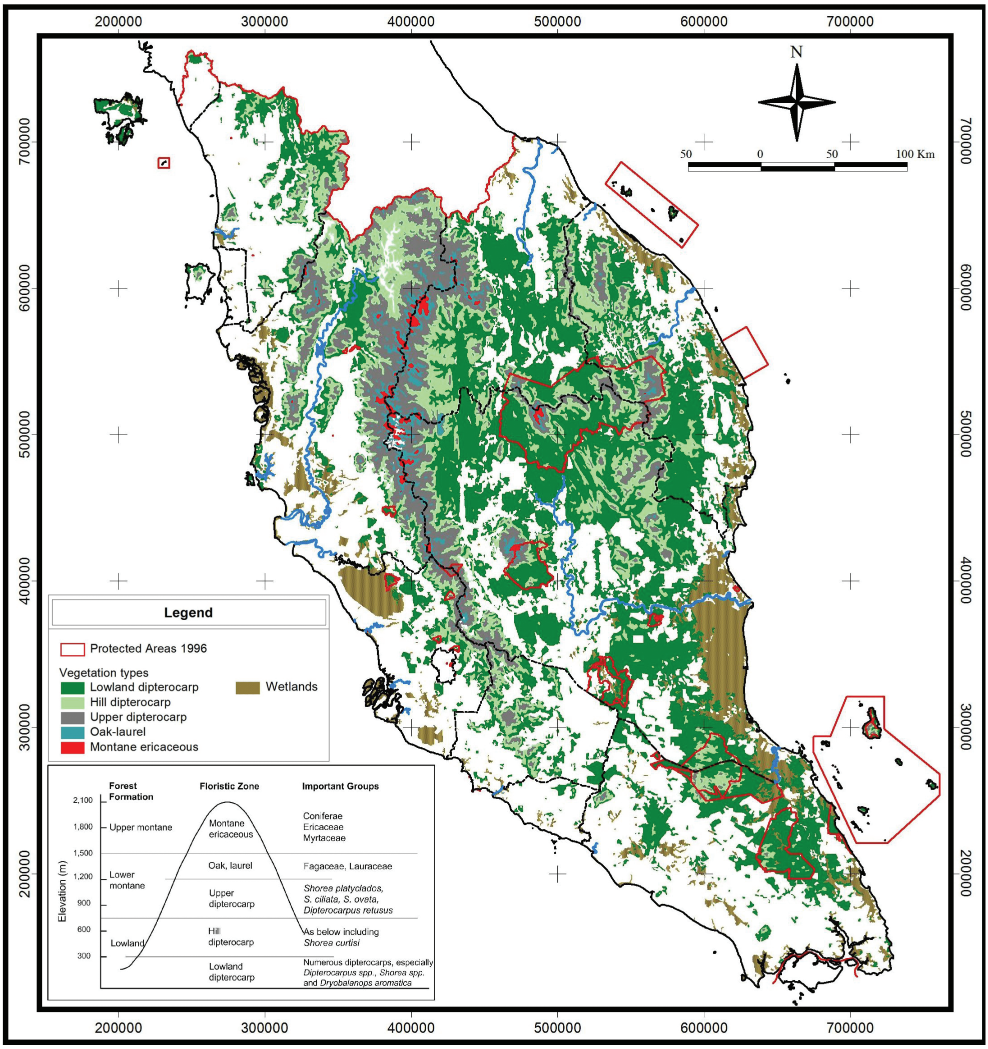
Climate	Soil water		Soils	Elevation		Forest formation	Vegetation types in Peninsular Malaysia
Ever-wet (perhumid)	Dry land		Zonal soils (mainly oxisols, ultisols)	Lowlands a. to 300 m b. to 800 m c. to 1200 m		Lowland evergreen rain forest	a. Lowland dipterocarp forests b. Hill dipterocarp forests c. Upper hill dipterocarp forests
				Mountains	(750) 1200–1500 m	Lower montane rain forest	Montane oak-laurel forests
					(600) 1500–2188 m	Upper montane rain forest	Montane ericaceous forests
			Podzolized sands	Mostly lowlands		Heath forest	Heath forest
			Limestone	Lowlands		Vegetation over limestone	Limestone vegetation
			Quartzite rocks	Lowlands to mountains		Vegetation over quartz	Vegetation over quartz
	Water table high (at least periodically)	Coastal salt-water		Coastal		Beach vegetation Mangrove forest Brackish water forest	Beach vegetation Mangrove forest Brackish water forest
		Inland fresh-water	Oligotropic peats	Mostly lowlands		Peat swamp forest	Peat swamp forest
			Eutropic (muck and mineral) soils	Lowlands		Freshwater swamp forest	Freshwater swamp forest
Seasonally dry	Moderate annual shortage		Mainly oxisols, ultisols	Lowlands		Semi-evergreen rain forest	White meranti-gerutu forest

The amount, type and persistence of water in soils (soil water) are another important influence in vegetation formation. In most soils, water is freely draining and no water logging occurs. When water accumulates periodically or permanently, specific vegetation types develop. Freshwater swamps develop in flat and low-lying areas of steams and rivers, where periodic flooding often occurs during the rainy seasons or where there is backing up from tidal influence. When water input is directly only from the rain in terrain that is flat, peat accumulates from the very slow decay of vegetation and peat swamp forest develops. Mangrove swamps occur in river mouths and deltas where fine marine alluvium accumulates with daily tidal influence. Further inland in river mouths under tidal influence with saltwater intrusion, the resulting vegetation is brackish water swamp. Finally, along the coast, where constant salt water and wave action shapes the beach and sea fronts, beach vegetation develops.

Forest types for Peninsular Malaysia were systematically described by Symington (1943) in the *Foresters' Manual of Dipterocarps* where he used the Dipterocarpaceae to define the different forest types. This was a very practical classification because dipterocarps dominate the upper emergent stratum of the lowland forests and are of primary importance to foresters. Wyatt-Smith (1963) in his *Manual of Malayan Silviculture for Inland Forest* further refined the classification based on the numerous ecological plots he set up. He further described many forest subtypes, in particular, for lowland forests where there is greater forestry interest, as well as other non-forest ecosystems. Based upon structure, physiognomy and floristic composition, forests can be more generally classified into forest formations irrespective of flora (Webb *et al*., 1967; Saw, 2004). This is a convenient classification because it allows for comparison of different forests found elsewhere. For the forests in Peninsular Malaysia, forest formation classification based on Wyatt-Smith's forest types is provided in Table 2. Forest profiles also provide a good diagrammatic representation of the structure of forests. Figure 1 provides the forest profiles of the four main lowland forest types. A generalised vegetation type map is provided in Map 1. This map provides vegetation-type distribution in Peninsular Malaysia based on elevation and wetlands for extant forests in 1990 (NRE, 2008) and the land areas of the major forest types based on the map are provided in Table 3.

Table 3. Generalised habitat types and extent of major forest types in Peninsular Malaysia in 1990 (NRE, 2008).

Vegetation type	**Altitudinal range (m)**	**Original extent**		**1990**	
		Million ha	**%**	**Million ha**	**%**
Lowland					
Wetlands	<300	1.187	9.0	0.955	7.2
Lowland dipterocarp	<300	9.160	69.5	3.739	28.4
Hill dipterocarp	300–750	1.475	11.2	1.441	10.9
Upper dipterocarp	750–1,200	1.099	8.3	1.094	8.3
Lower Montane	1,200–1,500	0.188	1.4	0.184	1.4
Upper Montane	>1,500	0.064	0.5	0.063	0.5
Total		13.173	100	7.476	56.7



Map 1. A generalised vegetation map for Peninsular Malaysia based on topography and wetlands (reproduced with permission from NRE, 2008).

Climatic Forest Vegetation Formations

Lowland evergreen rain forest

This is the main forest formation found in lowland areas of Peninsular Malaysia with perhumid (aseasonal) climate and mean diurnal temperatures ranging from 20°C to 34°C, annual rainfall of >2000 mm, and with no distinct dry season. The forest formation develops on well-drained soils not subject to water-logging. Typically soils are acidic, mainly oxisols and ultisols. Forests develop at elevations ranging from sea level to 1200 m.

This forest is conventionally regarded as having three tree layers, although the layers may grade into each other (Fig. 1) with (i) the upper layer of individual or grouped giant emergent trees sometimes as tall as > 70 m, some of the biggest trees have a clear bole of 30 m and reach 1.4 m diameter at breast height (dbh); (ii) the main stratum at about 24–36 m; and (iii) the lower layer with smaller, shade-tolerant trees and immature trees of the upper two layers. Ground vegetation is often sparse, and comprises mainly small trees, shrubs and understorey palms; herbs are scattered. Very few tree species are deciduous or semi-deciduous; the evergreen nature of the canopy predominates. Boles are usually cylindrical. Characteristic features of this formation are buttresses; cauliflory and ramiflory; and leaves that are often large and pinnate or variously dissected. Lianas (big woody climbers) are frequent. Shade and sun epiphytes are occasional to frequent.

Tree species diversity is among the richest in the world. Large ecological plot studies have enumerated 817 tree species ≥ 1 cm dbh per 50 ha in Pasoh FR, Negeri Sembilan (Manokaran *et al*., 2004) and in Sarawak the diversity is even greater with 1,171 species recorded in 52 ha in Lambir Hills NP (Plotkin *et al*., 2000). The family Dipterocarpaceae dominates the upper or emergent storey and can comprise 50% of the individuals of this storey.

In Peninsular Malaysia, based on changes in species composition that occur with altitude, Symington (1943) and Wyatt-Smith (1963) recognised a further three sub-forest types within the lowland evergreen rainforest formation, namely, lowland dipterocarp forests (0–300 m elevation, Plate 1A & 3A), hill dipterocarp forests (300–750 m) and upper hill dipterocarp forests (750–1200 m, Plate 1B). Symington (1943) included the upper hill dipterocarp forests in lower montane forest, however, here, following Whitmore (1984), it is included as part of the lowland rainforest formation. There is no sharp differentiation between these forest types; the main difference is the shift in the floristic composition of the dominants in the upper and main tree storeys. In coastal hills, the hill dipterocarp elements may occur at lower elevations. The more dominant species in the lowland dipterocarp forests include mainly the Dipterocarpaceae (*Anisoptera*, *Dipterocarpus*, *Dryobalanops*, *Hopea*, *Parashorea* and *Shorea*). Other common large trees are *Dyera costulata* (Apocynaceae), *Gluta* spp. (Anacardiaceae), *Heritiera* spp. (Malvaceae), *Intsia palembanica* (Leguminosae), *Koompassia malaccensis* (Leguminosae), *Palaquium* spp. (Sapotaceae) and *Sindora* spp. (Leguminosae). In some alluvial forest areas the gigantic *Koompassia excelsa* trees (Leguminosae) can be seen towering above the rest of the forest canopy. The main storey trees include members from Burseraceae, Guttiferae, Myristicaceae, Myrtaceae (particularly *Syzygium*) and Sapotaceae. In parts of the East Coast and Johor, *Dryobalanops aromatica* grows in gregarious groves on deeper soils edging sandstone formations. The understorey tree layer mainly comprises saplings of the upper storey and shrubs and climbers from such families as Annonaceae, Euphorbiaceae, Flacourtiaceae, Melastomataceae and Rubiaceae. Palms (Palmae) have a dominant presence in this stratum and species such as *Arenga westerhoutii*, *A. obtusifolia*, *Eugeissona tristis* (on low ridges) and *Oncosperma horridum* are common, while *Licuala*, *Iguanura* and *Pinanga* sometimes dominate the forest floor together with the non-climbing rattans, *Calamus* and *Daemonorops* spp. The dicotyledonous herbaceous flora is generally poor at lower elevations. However, monocotyledons, such as Zingiberaceae, occasionally Marantaceae and forest Cyperaceae (especially *Mapania*), *Orchidantha* (Lowiaceae) and *Tacca integrifolia* (Taccaceae) can be common in many forest areas. In more moist valleys and steep stream banks some genera such as *Argostemma* (Rubiaceae), *Begonia* (Begoniaceae), *Henckelia* (Gesneriaceae) and *Sonerila* (Melastomataceae) can become common.

In hill dipterocarp forests, many of the common lowland dipterocarp species are also

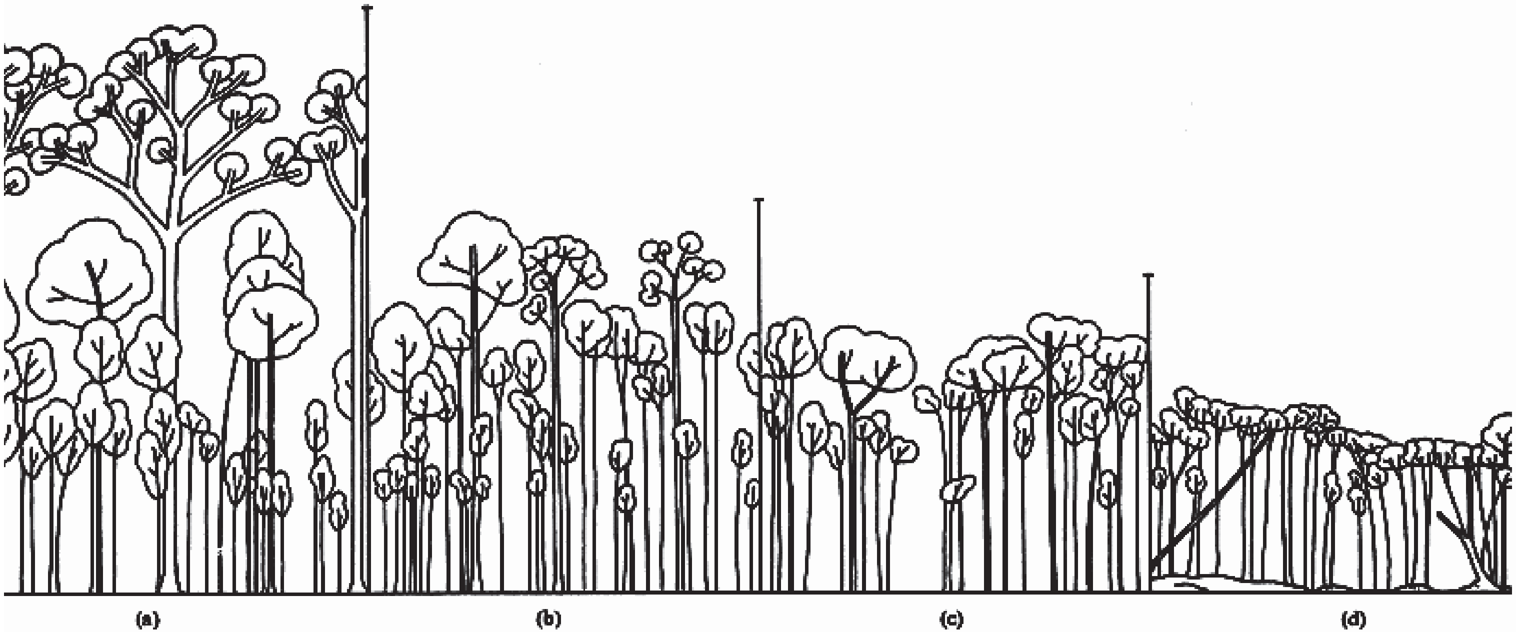


Figure 1. Profile diagrams of the major climatic forest formations in Peninsular Malaysia based on altitude (adapted from Wyatt-Smith, 1963). (a) Lowland dipterocarp forest, at 150 m a.s.l., Jengka FR, Pahang. (b) Upper hill dipterocarp forest, at 790 m a.s.l., G. Jerai, Kedah. (c) Lower montane oak-laurel forest, at 1,520 m a.s.l., G. Berembun, Cameron Highlands, Pahang. (d) Montane ericaceous forest at 1,830 m a.s.l., G. Berembun, Cameron Highlands. The first two, (a) and (b), belong to the lowland evergreen rain forest formation dominated by Dipterocarpaceae (open crowns), the formation incorporates many forest types of different stature and floristic composition but nevertheless exhibits three tree layers throughout its range from the lowland to the upper hill zone. The third section (c) shows the two tree layers of lower montane forest formation, whilst the last is of the single-layered upper montane forest formation.

found but here some species begin to disappear and are replaced by other species. The most common and dominant tree species is *Shorea curtisii* (Plate 1B) which tends to be gregarious and shows a distinct preference for ridges. It often grows in association with the understorey palm *Eugeissona tristis.* On ridges, except for *E. tristis*, the undergrowth vegetation becomes very sparse or non-existent (Wong *et al*., 1987).

In upper dipterocarp forests, the family Dipterocarpaceae that is a characteristic feature of the lowland and hill dipterocarp forests is represented by only a few species. The forest is often characterised by the presence of *Agathis borneensis* (Araucariaceae) and *Shorea platyclados*. Other tree species include *Calophyllum* spp. (Guttiferae), *Dipterocarpus costatus*, *D. retusus*, *Gluta* spp., *Shorea ciliata*, *S. ovata* and *S. submontana*. The characteristic tree fan palm, *Livistona speciosa* (Palmae, Plate 3B), is sometimes common on upper ridges of the Main Range. Under the cooler and wetter environment, the herbaceous flora becomes richer than at lower elevations. Similar families and genera that are found in the lowlands now become more common but their species, however, are different. Altitudinal zonation within the lowland rainforest formation has not been well-documented for understorey species but in some groups like the begonias, lowland species are replaced by others in hill and upper hill forest (Kiew, 2005).

***Semi-evergreen rain forest* (*White meranti-gerutu forest*)**

This forest type occurs in the extreme north-west of Peninsular Malaysia and covers Perlis, the islands of Langkawi (Plate 2A) and part of mainland Kedah north of G. Jerai, Bukit Perak and Padang Sanai, but excluding Bukit Perangin and Bukit Koh Mai hill mass.

The climate here has a distinct annual dry season of two to three months during December to March. While the total annual rainfall is between 2000 and 2300 mm, the average for January and February is below 50 mm. Soils are well-drained and not subject to water-logging, comprising various soil types, but mainly oxisols and ultisols. This forest type is found in the lowlands to about 300 m elevation.

Structurally, it is a strictly three-layered lowland dipterocarp forest. The upper canopy reaches a height of about 35 m, the main canopy about 24 m and the understorey about 12 m. It includes both evergreen and, in the upper canopy, deciduous trees in a mixture but with a definite tendency to gregarious occurrence. Deciduous trees may comprise up to one-third of the taller trees, although not all are necessarily leafless at the same time. Buttresses continue to be frequent in both evergreen and deciduous species. Bark tends to be thicker and rougher, and cauliflory and ramiflory are rarer. However, the forest is less dense, the upper canopy slightly lower, the form of trees poorer, and scramblers and woody climbers more frequent. Bamboos are present and are more common in areas that have been disturbed. Epiphytes are occasional to frequent and include many ferns and orchids.

The pronounced dry season is reflected in species composition, which predominantly includes Burmese and Thai species. The characteristic features of this forest type are the dominance of the white meranti group of *Shorea* while other *Shorea* groups are lacking. Common species include *Shorea assamica* subsp. *globifera*, *S. henryana*, *S. hypochra* and *S. roxburghii.* Other common and characteristic dipterocarps are *Anisoptera costata*, *Hopea ferrea*, *H. helferi*, *H. latifolia*, *Dipterocarpus baudii*, *D. dyeri*, *D. kerrii*, *D. grandiflora*, *D. obtusifolius*, *Parashorea stellata* and *Vatica cinerea*. Other common species include *Dillenia obovata* (Dilleniaceae), *Parkia timoriana* (Leguminosae), and *Pentace curtisii* (Malvaceae). Large trees of *Intsia palembanica* and *Koompassia malaccensis* are occasionally present. In areas that have been subjected to disturbance, *Schima*-bamboo forest develops, typified by *Schima wallichii* (Theaceae) and *Shorea roxburghii* and by the bamboos *Gigantochloa latifolia* and *G. ligulata* (Gramineae). Other common species in this forest include *Anisoptera costata*, *Cassia javanica* subsp. *nodosa* (Leguminosae), *Cratoxylum cochinchinense* (Hypericaceae), *Crypteronia paniculata* (Crypteroniaceae), *Derris microphylla* (Leguminosae), *Elaeocarpus robustus* (Elaeocarpaceae), *Garcinia hombroniana* (Guttiferae), *Ilex cymosa* (Aquifoliaceae), *Lagerstroemia floribunda* (Lythraceae), *Litsea grandis* (Lauraceae), *Parinari* spp. (Chrysobalanaceae), *Peltophorum dasyrhachis* (Leguminosae), *Syzygium gratum*, *Terminalia calamansanai* (Combretaceae), *Vatica cinerea* and *Vitex pinnata* (Labiatae). The herbaceous flora is generally very poor due to the drier climate.

***Lower montane forests* (*Montane oak-laurel forests*)**

There is an increase in cloudiness and rainfall as elevation increases. Temperature decreases with increasing elevation at a rate of about 0.65°C per 100 m but this varies from place to place, with season, time of day, water vapour content of the air and other factors. Cloud line is at about 1200 m. Lower montane forest develops on well-drained soils mainly at elevations between 1,000 and 1,500 m. Soil composition changes with elevation from that of the lowland forest, in general becoming more humic with available nutrients diminishing, especially where peat develops. The peat layer in lower montane forests is, however, generally thin.

In Peninsular Malaysia on mountain ranges such as those of the Main, Bintang, Keledang, Benom, Tahan and Terengganu Ranges, lower montane forest occurs just above the upper dipterocarp forest. It generally has two tree layers (Fig. 1) and reaches 15–33 m high, the upper canopy is fairly even with emergents absent. Trees have a relatively short bole, rarely exceeding 40–50 cm dbh, and are not strongly buttressed; cauliflory is uncommon, and the leaves are mainly medium-sized mesophylls, compound leaves are less frequent. Large lianas are not common, although smaller climbers may be found. Vascular epiphytes and ferns are abundant. Rattans, stemmed palms and tree ferns dominate the shrub layer and herbaceous plants are more abundant in the ground layer.

The flora is also as rich as the lowland forests. Kiew (1998), for example, listed over 900 species of seed plants from the mainly lower montane forest of Fraser's Hill, Pahang (Plate 2B). Although large emergent dipterocarps are not normally found in these forests, there are at least 11 dipterocarp species known from such elevations (species of *Dipterocarpus*, *Shorea* and *Vatica*). Other common characteristic species are representatives of the Fagaceae (*Castanopsis*, *Lithocarpus* and *Quercus*) and Lauraceae. Others include species of *Acer* (Sapindaceae), *Adinandra* (Theaceae), *Agathis*, *Calophyllum*, *Canarium*, *Dacrydium*, *Engelhardtia* (Juglandaceae), *Garcinia*, *Gordonia* (Theaceae), *Podocarpus* (Podocarpaceae), *Syzygium*, *Santiria* (Burseraceae) and *Toona* (Meliaceae). Species of climbers, e.g., *Aeschynanthus* (Gesneriaceae) and herbaceous plants, such as *Argostemma*, *Cyrtandra* (Gesneriaceae), *Filetia* (Acanthaceae), *Henckelia*, *Phyllagathis* (Melastomataceae), *Sonerila* and terrestrial orchids, *Calanthe*, *Hylophila*, *Liparis*, *Malaxis*, *Nephelaphyllum*, *Spathoglottis*, etc. (Orchidaceae), are very common. *Gingers*, such as *Alpinia*, *Amomum*, *Camptandra*, *Etlingera*, *Geostachys*, *Globba*, *Plagiostachys*, *Hornstedtia* and *Zingiber* (Zingiberaceae), are also common in this forest. The mountain fishtail palm, *Caryota maxima*, and the bamboo, *Schizostachyum grande* (Gramineae), are often associated with this forest at its lower elevations. On some mountain ridges, the endemic scrambling bamboo, *Maclurochloa montana* (Gramineae), may be common (Wong, 1995). Some palms may also be common, for example species of *Licuala*, *Iguanura*, *Nenga* and *Pinanga*.

***Upper montane forests* (*Montane ericaceous forests*)**

This formation develops above 1,500 m, above the cloud line, and experiences lower temperatures than does lower montane forest. Moisture is particularly high both from high rainfall, e.g., Cameron Highlands (Plate 3C) averages 2,650 mm annually (Wyatt-Smith, 1963), and frequent mist from clouds. Peaty soils begin to form above 1,000 m and high rainfall in this zone results in increased leaching of soils, podzolization, soil water-logging, and high acidity (pH 3–3.5). At the lower temperatures, biological activity and chemical weathering are also retarded. Soils here also tend to be shallower than those at lower elevations. Due to the *Massenerhebung* (mass elevation) effect, the upper montane forest develops at lower elevations on isolated peaks, e.g. G. Belumut (770 m), G. Ledang (753 m) and G. Jerai (915 m) and on such sites, lower montane forest may not be prominent.

The forest canopy is about 1.5–18 m high, flattened and compact, comprised of a single tree layer and the ground flora (Fig. 1). Emergent trees are usually absent. Tree trunks are often crooked and gnarled, often without buttresses. Leaves of trees and shrubs are often microphylls, pinnate leaves are very rare, and cauliflory is absent. Lianas are absent, bole climbers are very few, but in contrast the tree trunks are covered by bryophytes (mostly by leafy liverworts), lichens and vascular epiphytes, particularly orchids and ferns. The forest floor is thickly covered by mosses (for example *Sphagnum*) or by leafy liverworts.

The more common tree families in this vegetation type include Araliaceae, Araucariaceae, Clethraceae, Cunoniaceae, Ericaceae, Fagaceae, Lauraceae, Myrtaceae, Pentaphylacaceae, Podocarpaceae, Sapindaceae, Symplocaceae and Theaceae and the more common species include trees like *Dacrydium* (Podocarpaceae), *Daphniphyllum* (Daphniphyllaceae), *Eurya* (Theaceae), *Ficus* (Moraceae), *Gordonia* (Theaceae), *Ilex*, *Leptospermum* (Myrtaceae), *Lindera* (Lauraceae), *Lithocarpus*, *Melicope* (Rutaceae), *Podocarpus*, *Prunus* (Rosaceae), *Quercus*, *Syzygium*, *Schima*, *Ternstroemia* (Ternstroemiaceae), *Tristaniopsis* (Myrtaceae) and *Weinmannia fraxinea* (Cunoniaceae); climbers like *Crawfurdia* (Gentianaceae), *Frecynetia* (Pandanaceae), *Nepenthes* (Nepenthaceae) and *Rubus* (Rosaceae), and shrubs, such as species of *Rhododendron* and *Vaccinium* (Ericaceae), and herbs like *Argostemma* (Rubiaceae), *Ophiorrhiza* (Rubiaceae); *Piper* (Piperaceae), *Psychotria* (Rubiaceae) and terrestrial orchids, e.g., *Corybas* (Chua & Saw, 2001; Stone, 1981). Palms, although not diverse, in some areas dominate the forest floor, e.g., *Pinanga perakensis* and *P. polymorpha*, but rattan species are not common although a few species are upper montane forests specialists, e.g., *Calamus viridispinus*. Tree ferns (*Cyathea*, Cyatheaceae) are conspicuous. In some mountain areas, as in lower montane forests, some scrambling-bamboos, such as *Holttumochloa magica* (Gramineae), may dominate summits and ridges. Pandanaceae can also be very common, in Peninsular Malaysia for example, *Pandanus klossii*, a tree pandan is very much a component of the montane vegetation. Epiphytes too are particularly diverse in this habitat with ferns, including filmy ferns and species of Grammatidaceae and a great variety of orchid species. Within this forest formation, there is also a shorter facies on the more exposed ridges and crests called the elfin or mossy forest (Stone, 1981). Trees here are even shorter, more stunted and gnarled on exposed summits and ridges than in valleys and gullies where trees can be better formed and taller. However, species composition is about the same but with some shift in dominance of specific species (e.g., on Ulu Kali greater abundance of *Dacrydium comosum*, *Leptospermum* and *Nepenthes* and lacking *Pandanus klossii*).

Edaphic Vegetation Formations

Heath vegetation

Heath vegetation forms over sandstone or over raised sandy sea beaches, the latter known as permatang (Wyatt-Smith, 1963) and these old raised sandy sea beaches often run parallel to the sea coast (Plate 3D). The soils are derived from siliceous parent material that is inherently poor in bases, highly acidic, lacking buffering capacity due to a shortage of sesquioxides, and are also commonly coarsely textured. A thin peat layer may form above the podzolic soil. The soils become temporarily waterlogged after heavy rain. Rivers draining heath forests, e.g., Sg. Tahan, are tea-coloured owing to the presence of organic colloids. They are usually acidic (pH < 5.5) and with low oxygen content.

The heath forest is very simple in structure in comparison to the lowland forest formations. From aerial photographs, the flat canopy is highly distinctive owing to its pale tone and its very fine texture, the result of tree-crown structure and small leaf size. The main storey comprises large saplings and small poles in a tidy and orderly manner, often in dense stands. The main canopy is low and variable depending on the depth of the soil and availability of water, but in areas with more developed soils it can be over 25 m tall. It is also uniform and usually densely closed with no trace of layering. Single emergents may occur and usually indicate extreme site conditions. There are more trees with microphylls than mesophylls and many leaves are distinctly sclerophyllous. Trees of large diameter are rare; buttresses are smaller, but stilt roots more common. Deciduous species are absent. Big woody climbers (including climbing palms) are rare but slender, wiry, independent climbers are frequent. Epiphytes are frequent. Due to the very poor nutrient availability, myrmecophytes (ant plants) and insectivorous plants are abundant especially in the more open and stunted heath forest. The ground cover may have a bryophyte layer. In areas where soils are thin, shrub vegetation (padang) only a few metres tall dominates and on extremely shallow soils with constant water-logging, the vegetation consists of only sedges and low shrubs.

Species in heath forest are different from forests at a similar altitude with perhaps less than half the species in common. Although species diversity is much lower than in the lowland forests, it is nevertheless species rich. For example, in Sarawak where this forest type is called kerangas forests, Brunig (1974) recorded altogether 849 tree species, 133 shrubs, 96 herbs, 100 epiphytes and 55 lianas. In Peninsular Malaysia, the raised sandy coastal heath occurs mainly in the East Coast and in a small area on the West Coast in Perak at Tanjong Hantu FR, Dindings. Some of the larger and more common trees found in the coastal heath forests include *Gluta* spp., *Hopea griffithii*, *H. semicuneata*, *Irvingia malayana* (Irvingiaceae), *Madhuca utilis* (Sapotaceae), *Mesua ferrea* (Guttiferae), *Sindora echinocalyx* and *Shorea materialis.* Pole-sized trees include *Eurycoma longifolia* (Simaroubaceae), *Garcinia hombroniana*, *G. nigrolineata*, *Guioa* spp. (Sapindaceae), *Morella esculenta* (Myricaceae), *Syzygium* spp., *Tristaniopsis obovata* (Myrtaceae) and *Vitex pinnata*. Most of the heath vegetation on the East Coast has been destroyed by burning and open grazing. Only two small semi-natural stands remain at Jambu Bongkok FR and Menchali FR, Terengganu.

Heath forests in Peninsular Malaysia, associated with sandstone massifs on sandstone outcrops or plateaux are found on a number of coastal mountain outcrops, all of these are outside of the Main Range. These include mountain peaks in the Endau-Rompin State Park (Johor-Pahang boundary, Plate 4A), G. Ledang and G. Panti, Johor, and G. Jerai, Kedah, and the most prominent of all, the padang on G. Tahan, Pahang (Plate 4B), right in the heart of Peninsular Malaysia (Wong *et al*., 1987; Ridley, 1915; Soepadmo, 1971). In Terengganu, the sandstone massifs do not form plateaux but are common on some of the coastal hill ridges, e.g., Bt. Bauk. When the sandstone formation extends into higher elevations, the montane flora extends to lower elevations. In many respects, the upper montane and heath forest have many features of structure and physiognomy in common (Whitmore, 1984). Both have pole-sized trees, even forest canopy and similar species composition. These heath formations on sandstone massifs are more similar to those found in Borneo than those of the coastal sandy heath. There is also greater variation in the forest or vegetation structure than is seen in the coastal heath formation, mainly due to greater variations in soil depth and microhabitats of gullies, hillocks and ridges. In the Endau-Rompin Park, for example, common species found there include *Calophyllum* spp., *Cotylelobium lanceolatum* (Dipterocarpaceae), *Gluta aptera*, *Leptospermum javanica* and *Tristaniopsis merguensis*. Ground vegetation includes *Leucopogon malayanus* (Ericaceae), and the ferns *Dipteris conjugata* (Dipteridaceae), *Matonia pectinata* (Matoniaceae), *Gleichenia microphylla* (Gleicheniaceae) and *Sticherus* (Gleicheniaceae) (Wong *et al*., 1987). At about 1,500 to 2,000 m, the padang vegetation at G. Tahan comprises many species from the upper montane forest. On these padangs, there are many facies of heath ranging from windswept *Leptospermum* 30–60 cm tall on open areas of rock with crevices shallow soil, to 3–5 m tall in gullies, and up to 15 m along streams.

Finally, a particular facies of the sandstone-derived heath is the gregarious stands of *Livistona endauensis* found just edging the heath forest on the sandstone massif of Endau-Rompin (Wong *et al*., 1987) and coastal hill forest in the south Terengganu hills. These *Livistona* palm forests (Plate 4C) occur on ridges on better drained soils of the sandstone massif, but where they are found on the margins of the heath forest they occur in lower densities.

Limestone forests

In Peninsular Malaysia, limestone hills are mostly found in the north in Perlis, Kedah (including the Langkawi Islands), Kelantan, Terengganu, Pahang, Perak and Selangor (Chin, 1977). The southernmost one in Johor is so small that it is shaded by the tree canopy.

In Peninsular Malaysia, limestone outcrops occur in lowland areas. The limestone tower karsts have a diversity of habitats and soils. The structure and physiognomy of the vegetation on the limestone hill depends on the microhabitats and develops from a combination of the soils, substrate and climate (Plate 4D). Around the base of the hills and in ravines, the accumulated alluvial soils are base-rich and fertile and often will support large trees of the neighbouring forests up to 30 m tall; the rocky substrate near the base of the hill supports mainly pole-sized trees interspersed by larger trees. In sheltered areas, in a microhabitat of high humidity, lush herbaceous plants are common, e.g. aroids, gesneriads, begonias, balsams and terrestrial orchids. Further up the slope are the sheer cliffs, where soil accumulate in cracks and crevices and a few pole-sized trees, strangling figs and shrubs can root. On the summit, a deep mat of peat-like humus sometimes develops held together by tree roots which anchor them onto the limestone. The summit vegetation has a structure similar to that of heath forest. Trees are frequently microphyllous and cauliflory and ramiflory are rare (Wyatt-Smith, 1963).

Limestone communities comprise distinct floras. Over 1,200 species of plants have been recorded on the limestone hills of Peninsular Malaysia of which a large number are endemic (about 21% of the limestone flora) and a small proportion strictly calcicole (at least 125 species or about 10% of the limestone flora) thereby being restricted to limestone

(Chin, 1977). Common trees and shrubs include *Cleidion spiciflorum* and species of *Bridelia* and *Cleistanthus* (Euphorbiaceae), *Callicarpa angustifolia* (Labiatae), *Dehaasia pauciflora* (Lauraceae), *Pentaspadon curtisii* (Anacardiaceae), *Podocarpus polystachyus* and species of *Buxus* (Buxaceae), *Diospyros* (Ebenaceae), *Ficus* and *Schefflera* (Araliaceae), etc. Dipterocarps are uncommon on limestone hills, although *Shorea glauca* and *Hopea bilitonensis* have been recorded. The most characteristic plants of the limestone vegetation are the small herbs and rock-face species e.g., species of *Impatiens* (Balsaminaceae) and Gesneriaceae (*Chirita*, *Epithema*, *Monophyllaea* and *Paraboea*). *Pandanus* species are also common. Orchids are particularly rich on limestone hills, both as epiphytes and as calcicoles, such as *Paphiopedilum niveum*, and species of *Bulbophyllum*, *Calanthe* and *Cleisostoma*. In Langkawi Island, Perlis and Perak, the strictly calcicole cycad, *Cycas clivicola* (Cycadaceae), is common on the summit clinging precariously in crevices of cliffs. On Kelantan limestone, *Cycas macrocarpa* is common on hills, where it occurs mainly in crevices where there is accumulation of soil. A number of palm species are restricted to limestone hills; these include two species of *Maxburretia*, *M. rupicola* (endemic to the Bt. Takun and Batu Caves, Selangor) and *M. graciles* (on limestone hills of Langkawi Island and Perlis), and *Licuala kingiana* (Perak). Following disturbance, the palms *Arenga westerhoutii* and *Caryota mitis* often dominate the lower slopes of the limestone hills.

Beach vegetation

Beach vegetation forms above the high tide zone (Plate 5A). Two kinds of beach vegetation are recognised. One is on accreting coasts where new sand is deposited and plant cover often consists of low herbaceous creeping plants (the pes-caprae association so-called because *Ipomoea pes-caprae*, Convolvulaceae is particularly common). The other is inland where shrubs give way to a tree community with simple canopy structure of generally medium-sized trees, mainly with mesophylls. Here climbers are common but large lianas are rare, and herbs are generally lacking.

Along the accreting sandy beaches *Casuarina equisetifolia* (Casuarinaceae) is the predominant species succeeded by *Calophyllum inophyllum*, *Cycas litoralis*, *Dendrolobium umbellatum*, *Erythrina variegata* and *Peltophorum pterocarpum* (Leguminosae), *Podocarpus polystachyus*, *Syzygium grande*, *Pandanus odoratissimus*, *Pouteria obovata* (Sapotaceae), *Scaevola taccada* (Goodeniaceae) and *Terminalia catappa* (Combretaceae). Between the tree belt and the high tide mark, there is usually a narrow herbaceous community of *Canavalia cathartica* and *Vigna marina* (Leguminosae), *Cyperus stoloniferus* and *Remirea maritime* (Cyperaceae), *Euphorbia atoto* (Euphorbiaceae), *Ipomoea pes-caprae* subsp. *brasiliensis*, *Ischaemum muticum* and *Spinifex littoreus* (Gramineae) with *Vitex trifolia* subsp. *littoralis* as a common creeping shrub and scrambler (Wyatt-Smith, 1963).

Along the receding and gravel beaches, notably on the West Coast of Peninsular Malaysia, the herbaceous community and usually also the *Casuarina equisetifolia* fringe are absent. In these cases, the littoral or strand forest consists of a narrow, often single belt of trees which includes *Barringtonia asiatica* (Lecythidaceae), *Calophyllum inophyllum*, *Cerbera manghas* (Apocynaceae), *Dendrolobium umbellatum*, *Guettarda speciosa* (Rubiaceae), *Hernandia nymphiifolia* (Hernandiaceae), *Hibiscus tiliaceus* and *Thespesia populnea* (Malvaceae), *Pongamia pinnata* and *Sophora tomentosa* (Leguminosae), *Scaevola taccada*, *Syzygium grande* and *Terminalia catappa*. Herbaceous plants such as *Tacca leontopetaloides* and *Crinum asiaticum* (Amaryllidaceae) can be quite common.

The beach vegetation of small coastal islands is no different from those found on the mainland. However, many of the smaller islands tend to be very rocky and the absence of sandy beaches results in the absence of sandy beach vegetation (Wyatt-Smith, 1953; Corner, 1985). Inland on larger islands, relics of the original vegetation remain but sometimes a few species may dominate due to lack of competition. Overall species diversity is reduced compared to the mainland vegetation. Species such as *Pisonia grandis* (Nyctaginaceae) and many strangling *Ficus* are particularly common on these islands (Corner, 1985). On some of the East Coast islands, on the side facing the east exposed to the north-east monsoon, wind-pruned vegetation is common where the annual strong winds shear off the vegetation at the beach fronts. Sometimes, the plants possess a peculiar oblique, condensed, one-sided habit, often gnarled, all about the same height, very short, sometimes less than a metre near the beach and progressively taller inland, for example on Redang Island, Terengganu.

Marine alluvial (*mangrove*) *swamp forests*

The mangrove swamp forest grows on muddy shores, lagoons and muddy estuaries of tidal rivers (Plate 6A). Mangroves fringe a large extent of the West Coast coastline especially in Perak, of which the Matang mangrove is the largest, followed by those in Johor and Selangor. On the East Coast, the annual monsoons prevent development of large areas of muddy shores in many of the rivers draining out to the South China Sea. Mangroves here are confined to sheltered river mouths and estuaries.

Soils range from sandy to fine, newly deposited silt and heavy clay. The forest structure is very simple, 5–25 m tall in height depending on the community with a comparatively even and unbroken canopy with a very species-poor understorey layer. Trees with special breathing roots (pneumatophores) or stilt roots are common. Leaves are mesophylls, cauliflory and ramiflory are absent, viviparous germination is common. Large lianas and climbers are absent.

About 60 tree species are recorded from the mangrove swamps of Peninsular Malaysia, making it one of the richest in the world (Saenger *et al*., 1983) in part due to the range of ecotypes in a complex zonation of different communities (Watson, 1928). Thus *Ceriops decandra* (Rhizophoraceae), *Avicennia alba* and *A. rumphiana* (Avicenniaceae) are pioneer species on stiff clay and in the less sheltered parts of the river; *Avicennia marina* and *Bruguiera cylindrica* (Rhizophoraceae) form almost pure stands on stiff clay; *Rhizophora apiculata* (Rhizophoraceae) lines the banks of creeks in almost pure stands with a mix of *Bruguiera parviflora* and *Rhizophora mucronata* while species further inland include *Brownlowia argentata* (Malvaceae), *Bruguiera sexangula*, *B. gymnorhiza* (Plate 6B), *B. hainesii*, *Heritiera littoralis* (Malvaceae), *Intsia bijuga* (Leguminosae), *Xylocarpus granatum* (Meliaceae), *Oncosperma tigillarium* (a palm that often forms groves) and the ferns *Acrostichum aureum* and *A. speciosum* (Parkeriaceae).

Brackish water vegetation

On the inland edge of mangrove and the upper tidal limit of estuaries, brackish water vegetation forms. The soils here are predominantly heavy alluvial clay. Structure is very simple consisting of stands of *Nypa* palm and associated communities. Pure stands of *Nypa fruticans* (Palmae) are common in Peninsular Malaysia, while *Phoenix paludosa* (Palmae) grows in some areas just behind the *Nypa* stands on drier ground that is still subject to periodic flooding (Plate 6C). In some areas, *Livistona saribus* (Palmae) grows in almost pure stands just behind the *Nypa* stand (e.g. in Kemaman, Terengganu) and also extends further inland into freshwater swamp habitats. Other species include *Artocarpus teysmannii* (Moraceae), *Brownlowia argentata*, *Cerbera odollam*, *Excoecaria agallocha* (Euphorbiaceae), *Ficus microcarpa*, *Intsia bijuga* and the palm *Oncosperma tigillarium*. The large climbing rattan, *Calamus erinaceus*, is also a distinctive feature of this vegetation type.

As for all types of swamp forest, the brackish water habitat in recent years is highly threatened by development especially in areas that are close to urban centres. For example, the *Phoenix paludosa* stands on both the East (Paka, Terengganu) and West Coasts (Yan, Kedah and Butterworth, Penang) have mostly been destroyed.

Peat swamp forests

The peat swamp forests develop in areas that are permanently water-logged and where the incoming water is from rain, i.e., it is mineral deficient. The peat soils can vary in depth from a few to about seven or more metres deep and because of the waterlogged conditions they are anaerobic. The water in drainage channels is almost black in appearance and is always very acidic. Well-developed peat swamp is usually characterised by a domed or convex surface, consequently flooding by adjacent rivers is only restricted to the margins. The peat consists of dark reddish-brown to black, semi-decomposed leaves, branches, twigs and tree trunks. The underlying material over which peat deposits accumulate consists of either sulphidic marine clay or riverine alluvium and sand. Peat swamp forests are found in SE Pahang, Johor, Selangor and Perak and are now highly threatened by development, particularly by draining and clearing for pineapple or oil palm plantations.

These forests have a three-layered tree structure with emergent, middle and understorey trees. Stem density is high with smaller sized trees compared to those of the lowland dipterocarp forests. Trees often have stilt and aerial roots, while some have spreading buttresses. Leaves are mainly mesophylls, pinnate leaves are occasional. The ground vegetation is poor often consisting of seedlings of the trees but including *Pandanus* species and groves of the palm, *Eleiodoxa conferta* (Palmae). Large climbers are rare.

Peat swamp forests in Sabah and Sarawak are more developed than in Peninsular Malaysia. In Sarawak, there exists a catena of forests from the edge to the centre of each peat swamp. Anderson (1963, 1964) divided these into six types that are distinct in structure and physiognomy. Such differentiation is not very obvious in the peat swamp forests of Peninsular Malaysia.

Only relatively few species are specific to the peat swamp forest habitat, e.g., *Gonystylus bancanus* (Thymelaeaceae), *Shorea platycarpa*, *S. uliginosa* and the sealing wax palm, *Cyrtostachys renda* (Palmae). However, a large number of species from the inland forests extend into this habitat. This may be due to the relatively young age of the peat swamp forests, less than 11,000 years (Whitmore, 1984). Anderson (1963) recorded only 317 species of flowering plants in this forest type in Sarawak and Brunei. Among the tree species found in the peat swamp forests in Peninsular Malaysia are *Austrobuxus nitidus* (Picrodendraceae), *Blumeodendron subrotundifolium* (Euphorbiaceae), *Bhesa paniculata* (Celastraceae), *Campnosperma auriculatum* and *C. coriaceum* (Anacardiaceae), *Cratoxylum arborescens*, *Diospyros maingayi*, *Durio carinatus* (Malvaceae), *Ficus* spp., *Gonystylus bancanus*, *Koompassia malaccensis*, *Lophopetalum multinervium* (Celastraceae), *Litsea grandis*, *Myristica gigantea* (Myristicaceae), *Nephelium maingayi* (Sapindaceae), *Parastemon urophyllus* (Chrysobalanaceae), *Pouteria maingayi*, *Polyalthia glauca* and *P. sclerophylla* (Annonaceae), *Pometia pinnata* (Sapindaceae), *Shorea platycarpa*, *S. teysmaniana*, *S. uliginosa* and *Stemonurus scorpioides* (Icacinaceae) (Ng & Shamsudin 2001; Wyatt-Smith1963). In disturbed areas or logged-over areas, *Macaranga pruinosa* (Euphorbiaceae) and *Cratoxylum arborescens* can form pure stands. Palms, such as *Cyrtostachys renda* and *Eleiodoxa conferta*, can be common. The only dominant rattan found in this swamp forest is *Korthalsia flagellaris.*

Freshwater swamp forest

Freshwater swamp forest develops over soil that is subject to regular to occasional flooding from river systems. Flooding can occur each day or monthly instead of seasonally, by the backing-up of river water at high tide. The fluctuating water level thus allows periodic drying of the soil surface. The alluvial soils are overlain by a few centimetres of peat or muck but are inundated by mineral-rich freshwater of fairly high pH (6 and above). Such swamps occur mainly at sea level but can occur at higher elevations where conditions for swamp formation in river systems are suitable. The freshwater swamp forest is found in flood plains throughout the country that are subjected to regular flooding. Such habitats are similar to the brackish swamp forest in being highly threatened by development in recent years. Much of the freshwater swamp forest of south Johor and Singapore described by Corner (1978) is already destroyed due to development and conversion to oil palm plantations. The freshwater swamp forest of Lumut district, Perak, that contains some of the rarest trees (e.g., *Dipterocarpus semivestitus* and *Vatica flavida*) are now confined to remnant areas.

The freshwater swamp forest is a heterogeneous community, varying from low scrub with scattered 20–30 m tall trees to a forest similar to that of peat swamp forest. Where flooding is brief, the forest approaches the lowland dipterocarp forest in structure and composition. Stilt roots, knee roots and plank-like sinuous buttresses are common features of some tree species found here. Often such forest may be dominated by thickets of the spiny clustered palm *Eleiodoxa conferta*. In the permanently flooded areas submerged plants, like species of *Cryptocoryne* (Araceae), may form colonies.

Common tree species include *Alstonia spatulata* (Apocynaceae), *Artocarpus teysmannii*, *Calophyllum* spp., *Campnosperma* spp. (in particular *C. coriaceum*), *Castanopsis* spp., *Dialium* spp., *Dillenia reticulata*, *Dipterocarpus costulatus*, *D. oblongifolius*, *Dryobalanops oblongifolia*, *Ficus* spp., *Hopea mengarawan*, *Ilex cymosa*, *Intsia palembanica*, *Koompassia malaccensis*, *Lophopetalum* spp., *Madhuca* spp., *Gluta* spp., *Mezzettia* spp. (Annonaceae), *Ochreinauclea maingayi* (Rubiaceae), *Palaquium* spp., *Parastemon urophyllus*, *Pometia pinnata*, *Shorea hemsleyana*, *S. macrantha*, *S. platycarpa*, *S. bentongensis*, *Sindora wallichii*, *Syzygium* spp., *Tristaniopsis* spp., *Vatica lobata* and *V. pauciflora*. Often a number of large palms are also common, most noticeable are the two endemic palms, *Pholidocarpus kingianus* and *P. macrocarpus* (Plate 6D). In some areas, there can be colonies of *Livistona saribus* in swamp vegetation edging rivers.

Corner (1978) listed a number of belts of swamp vegetation communities, each with their dominant species. He used the vernacular names of the dominant group so that the putat-belt, inland just behind the brackish water forests in the tidal freshwater zone, is characterized by *Barringtonia conoidea*, *B. racemosa* and *B. acutangula*. It sometimes forms thickets on incipient mud-banks. The rassau-belt follows behind the putat-belt as the mud band widens and is dominated by the rassau tree pandan, *Pandanus helicopus* (Plate 5B) together with *Gluta velutina.* The belt with *Polyalthia sclerophylla* (mempisang), *Elaeocarpus macrocerus* and *Horsfieldia irya* (Myristicaceae) follows behind the rassau-belt as the first formation on the stabilised mud banks that are limited to the freshwater tidal region. The jejawi-belt, characterized by *Ficus microcarpa* (jejawi), grows behind the rassau and mempisang belts and are chiefly in the lower and variably brackish region. The pelawan belt is dominated by *Tristaniopsis whiteana* (pelawan) that occupies the firm, raised river-banks especially in the freshwater tidal zone. Finally the banks of tributaries under the forest canopy are saraca-streams fringed by *Saraca cauliflora* (Leguminosae) with *Pentaspadon motleyi* (Anacardiaceae) and various dipterocarps.

Riparian vegetation

The riparian community is rather complex because it is a mix of plant associations depending on the size of the stream or river, flow of the water current, and the substrate of the stream or river bank. The nypa-rassau-saraca stream communities have been described above. The upper reaches of streams in steep terrain are often rocky. The periodic flooding of the banks of larger streams by the fast flowing water current washes away loose soil and even large stones, leaving behind large boulders breaking the torrential flow. As the stream drains to gentler slopes, it widens with the greater volume of water and a mix of rocky boulder banks and banks with sandy or pebbled deposits create different habitats. Often larger streams and rivers cut into the terrain resulting in earth banks, an additional habitat for plants. As the stream meanders into yet gentler terrain, some areas may even flow into fresh water swamps as described earlier. Ultimately, the river joins the larger river system that drains into the other brackish water and mangrove swamps and eventually into the sea.

Rheophytes, plant species that grow in and by torrential rivers, show remarkable similarities in habit and morphology that allow them to adapt to the strong water current (van Steenis, 1981). For example, they are firmly anchored to the river bed, bank, rocks or rapids by strong deep roots; their leaves are either small or substantially dissected or frequently narrow to linear, and they produce basal shoots easily and often have a tufted habit. Examples of rheophytes that grow on the rocky banks and spits of the Sg. Tahan, Pahang, include *Aglaia yzermannii* (Meliaceae), *Antidesma salicinum* (Phyllanthaceae), *Calophyllum rupicola*, *Dysoxylum angustifolium* (Meliaceae), *Ficus pyroliformis* and *Homonoia riparia* (Euphorbiaceae) (Whitmore, 1984). *Piptospatha* (Araceae) that grows on rocks in torrential saraca streams is an example of an herbaceous rheophyte.

A common river bank dipterocarp is the neram (*Dipterocarpus oblongifolius*), a species restricted mainly to the rivers east of the Main Range (Corner, 1978). Its adaptation to the riparian environment is seen in its water-dispersed fruits, and the seedlings and saplings that have narrow, linear leaves whereas the adult trees have the typical elliptic leaves of other *Dipterocarpus*. Due to the lower canopy and moister environment, epiphytes are particularly rich on river-bank trees, for example, Henderson (1935) recorded at least 87 species of epiphytes (mostly orchids and ferns) on a single tree of *D. oblongifolius.*

Other common riverine species include *Saraca cauliflora*, *S. declinata*, *Tristaniopsis whiteana*, species of Meliaceae, *Syzygium salictoides*, *Neonauclea pallida* subsp. *malaccensis* (Rubiaceae), and herbs such as, *Croton caudatus*, *C. rheophyticus* (Euphorbiaceae) and *Hedyotis rivalis* (Rubiaceae), and on earth banks, species of Araceae, such as, *Schismatoglottis* and *Rhaphidophora beccarii*. In many places the pandan, *Pandanus monotheca*, and the fern, *Dipteris lobbiana* (Dipteridaceae) line the water's edge (Plate 7A) in both shady and more exposed areas (Wong *et al*., 1987). Submerged in the slow-flowing streams, *Cryptocoryne* (Araceae) and *Barclaya* (Nymphaeaceae) species sometimes form carpets on the stream bed.

Finally a number of herbaceous species specialise in growing on the boulders as lithophytes. Some of these are actually delicate plants that cannot survive the flooding and water currents. They, however, grow on the rock face facing away from the stream flow or on sheltered rock faces high above the regular flood level (Kiew, 2005). Many species of Begoniaceae, Gesneriaceae and some ferns are such specialists.

Other Aquatic Vegetation

Aquatic vegetation in natural lakes

There are only two large natural lakes in Peninsular Malaysia, Tasik Bera and Tasik Chini both in Pahang. Tasik Bera is the larger of the two lakes covering 6,150 ha while Tasik Chini has only 902 ha of open water and swamp (Plate 7B). Both these lakes are connected to the Sg. Pahang and experience counter flows into the lakes during high waters of the Sg. Pahang. In recent years, large parts of the lakes dry up completely during dry periods (Rafidah *et al.*, 2010).

The aquatic vegetation of both lakes is very similar. Tasik Bera is composed of four major habitat types (Lim *et al*., 1982; Rafidah *et al*., 2010): (i) the open water with floating *Utricularia punctata* (Lentibulariaceae), the submerged *Blyxa aubertii* var. *echinosperma* (Hydrocharitaceae) and *Hydrilla verticillata* (Hydrocharitaceae) and around the margin by *Nymphoides indica* (Menyanthaceae) and *Nymphaea pubescens* (Nymphaeaceae) that root in the mud but have floating leaves; (ii) the *Lepironia articulata* (Cyperaceae) reed swamp with *Eleocharis ochrostachys* (Cyperaceae) and other sedges; (iii) the rassau fringe of *Pandanus helicopus* that dominates the lake and stream margins; and (iv) the *Syzygium* swamp-forest stands that are basically freshwater swamp.

In the past, Tasik Chini was subjected to great water level changes that favoured lotus, *Nelumbo nucifera* (Nelumbonaceae) for which the lake became famous but the aquatic flora of Tasik Chini has altered greatly since 1995, when a dam was built across Sg. Chini effectively preventing changes in water level. This led to the drowning of the *Syzygium* swamp-forest and the drastic decline of the lotus population (Gregory, 2006). Then the S American water weed, *Cabomba furcata* (Cabombaceae), was introduced and with the sewage effluent from resort development it has proliferated out of control at the expense of native species like *Utricularia* (Chew & Siti-Munirah, 2010).

Seagrass beds

Seagrasses are the only flowering plant group that live in seawater. They are, however, only found in sheltered coastal waters in association with shallow inter-tidal pools, semi-enclosed lagoons, coral reef flats and also sub-tidal zones (Japar Sidik *et al*., 2006; Phang, 2000; den Hartog, 1970). The inter-tidal seagrass community is not entirely submerged, but is inundated twice daily with the rise of the tides. In general, coastal areas between mangroves and corals (from the low tide level to the coral fringe) form the habitat for seagrasses. There are 14 species recorded for Peninsular Malaysia (Japar Sidik *et al*., 2006; Phang, 2000) in three families: Cymodoceaceae (*Cymodocea*, *Halodule*, *Syringodium*, and *Thalassodendron*); Hydocharitaceae (*Enhalus*, *Halophila*, and *Thalassia*); and Ruppiaceae (*Ruppia*). *Ruppia* and *Thalassodendron* are less common because they are inland species fringing into brackish waters. Japar Sidik *et al*. (2006) reported 78 seagrass beds scattered throughout Malaysia. The most important areas are found off the south and east coasts of Peninsular Malaysia where the coast is least urbanised.

Other Vegetation Types

Quartzite vegetation

About two-fifths of Peninsular Malaysia consists of granitic rocks (Tjia, 1988). Within the granitic terrain, quartzite ridges occasionally protrude as exposed sheer cliffs. On such quartz dykes and quartzite outcrops, mineral soil is often very thin and nutrient poor, soil moisture is extremely low and dry humus or peat generally develops over a shallowly sandy soil. The largest is the Klang Gates quartz dyke (Plate 8B), 16 km long and jutting some 200 m above the surrounding plain northeast of Kuala Lumpur (Tjia, 1988).

The vegetation structure varies considerably, depending on the habitat. At its base, the vegetation is that of the lowland evergreen rainforest formation and associated community. On exposed rocks, the vegetation comprises mainly shrubs, sedges and grasses. In sheltered valleys and shaded gullies where soils are deeper pole-sized trees can be common. The dominant plants on the exposed ridge of Klang Gates include shrubs and trees of *Aleisanthia rupestris* (Rubiaceae), *Baeckea frutescens* (Myrtaceae, Plate 8C), *Eurycoma longifolia*, *Ficus deltoidea*, *Ilex praetermissa*, *Rhodoleia championii* (Hamamelidaceae), and the endemic grass *Eulalia milsumii* (Gramineae). In shaded sites, often under trees, ferns such as *Davallia denticulata* and *D. heterophylla* (Davalliaceae), *Oleandra pistillaris* (Oleandraceae) and *Syngramma dayi* (Adiantaceae) may be common (Kiew, 1982). Other interesting plants include *Acrymia ajugiflora* (Labiatae), *Parastemon urophyllus* (Chrysobalanaceae), *Begonia sinuata* (Begoniaceae), *Henckelia primulina*, *Capparis versicolor* (Capparaceae), *Vaccinium bancanum* (Ericaceae), *Fagraea auriculata* and *Rhododendron longiflorum*. In total, Kiew (1982) recorded 175 species on the Klang Gates, of which, 6 species are endemic to this ridge.

Disturbed vegetation and regenerated forests

Primary or virgin forest areas that remain relatively undisturbed are areas under National/ State Parks and Wildlife and Bird Sanctuaries. These areas constitute only about 6.8% of the total land area in Peninsular Malaysia (Table 1). Forest areas under the jurisdiction of the Forest Department are managed under a sustainable management system. In Peninsular Malaysia, forest areas are designated as protection or production forests. Production forest in Peninsular Malaysia constitutes about 2.8 million ha (about 21.3% of the land area in Peninsular Malaysia or just over half the forested area) and most of these forest areas have since been logged at least once (Chin *et al*., 1997). The protection forest covers about 1.9 million ha or about 14.4% of land area (Table 1).

The lowland dipterocarp forests have suffered the greatest decline from land clearing activities. Their original range covered about 69.4% of the total land area in Peninsular

Malaysia, but is now reduced to 28.3%, lost mainly to agriculture, mining and urbanisation. Forest areas in Peninsular Malaysia are now confined mainly to land above 300 m elevation. With production forest in the hills, the more difficult terrain has resulted in more challenging forest management conditions. During logging operations there is greater damage resulting in over 50% of the trees being uprooted or with broken trunks, lopped crowns, broken branches, gashes along trunks or torn roots (Wan Mohd. Shukri *et al*., 2005, 2006, 2007). This has resulted in second growth forests being highly variable and the stands skewed towards non-dipterocarps (Appanah, 2000). Studies now indicate that recovery of the second growth after logging is poor and not as predicted by the Selective Management System (SMS). The clumped distribution of dipterocarps, e.g., *Shorea curtisii* and *S. platyclados*, results in greater damage in spite of higher dbh limits put on dipterocarp extraction. Subsequently in the ensuing harvesting cycle, there will be a shift in tree composition towards non-dipterocarps, and some post-felling data indicate that *Syzygium*, *Swintonia* (Anacardiaceae), *Elateriospermum tapos* (Euphorbiaceae), species of Myristicaceae and Lauraceae will dominate the forest making up to over 30% of the species composition (Wan Mohd. Shukri *et al*., 2005, 2006, 2007). Bamboos (*Gigantochloa* and *Dendrocalamus*) and some palms (e.g., *Arenga obtusifolia*, *A. westerhoutii*, *Caryota mitis* and *Eugeissona tristis*) may become dominant in some sites and may hinder tree regeneration. In addition, Ashton (2008) further suggested that some of the slow growing tree species will not reproduce sufficiently to sustain numbers being cut before reaching reproductive size. This will result or has resulted in a decline in the heavy hardwoods in the long-term in these managed forests. It is likely that in the future primary indigenous vegetation types will exist only in the totally protected areas.

Logging also increases the extent of the forest margin resulting in an explosion of *Macaranga* within about ten years after logging (Primack & Lee, 1991), as well as the large forest herbs like wild bananas, *Musa* species (Musaceae), and gingers, such as *Etlingera* species. They persist until shaded out by regenerating forest.

At the other extreme is open ground, the result of clear-felling, shifting cultivation or abandoned cultivated land or tin-mining. Here conditions for seedling establishment are extremely harsh with full exposure to light, heat and water stress (unless the area is water-logged) and nutrient-poor soil, because without a protective layer of leaf litter or vegetation, the top soil and nutrients will quickly be eroded away by heavy rain. The first plants to invade these areas are herbaceous weed species, such as grasses (Gramineae), *Hedyotis* or *Spermacoce* species (Rubiaceae), *Hyptis* species (Labiatae) or species of Compositae, and in water-logged areas, sedges (Cyperaceae) and *Ludwigia* species (Onagraceae). Over the years, their roots stabilise the soil and the dying leaves and stems increase soil fertility and gradually they will be followed by a succession of woody shrubs, like senduduk *Melastoma malabathricum* (Melastomataceae), *Chromolena odorata* (Compositae), *Lantana camara* and *Stachytarpeta* species (Verbenaceae), which if left undisturbed will after about 15 years be succeeded by secondary forest or belukar species. Belukar species are pioneer tree species that are light demanding, fast growing and have low canopy height compared with forest (hutan). Species diversity is very low and almost none of these species can grow in primary forest. Belukar (jungle) is dense with thick undergrowth and a tangle of many climbers compared with the open undergrowth of primary forest. Common belukar tree species include species of *Adinandra*, *Alstonia*, *Archidendron* (Leguminosae), *Eurya*, *Ficus*, *Glochidion* (Euphorbiaceae), *Macaranga*, *Mallotus* (Euphorbiaceae), *Neolamarckia* (Rubiaceae) and *Trema* (Ulmaceae). Aggressive climbers include species of *Uncaria* (Rubiaceae), *Merremia* (Convolvulaceae) and the exotic *Mikania* species (Compositae). Once belukar forms a complete canopy, understorey conditions are ameliorated by shade and increasing soil fertility as a litter layer develops. Provided the belukar is sufficiently close to primary forest, forest trees can invade and after a further 40 to 50 years, they will shade out the belukar species and the forest will gradually return to its original state.

However, frequently this succession is interrupted, for example, by fire, or is halted, for example, by grazing. Fire encourages the growth of lalang, *Imperata cylindrica* (Gramineae), which can survive burning because its rhizome is deeply rooted. In dry weather its leaves dry and are easily burnt and so the lalang field is established as a fire climax. Grazing and grass-cutting along roadsides also prevents the development of the shrub layer and favours the spread of creeping grasses such as *Axonopus compressus* (Gramineae) and other low-growing herbs. *Dicranopteris linearis* (Gleicheniaceae) thickets can also halt succession because they blanket that ground and so prevent the establishment of light-demanding shrubs.

Many of the weedy herbs and shrubs are exotic species that have, in most cases, been introduced accidentally or are garden escapes, a process that continues today (Kiew, 2009). However, not only do these exotic species form only a small fraction of the flora (probably less than 5% of species) but in almost all cases they are unable to compete with forest species and cannot gain a foothold in forest. Their continued existence therefore depends on human disturbance.

Conclusion

The forests and other natural vegetation of Peninsular Malaysia are disappearing at an unprecedented rate. The conservation of these habitats should be given greater attention and priority. Although Peninsular Malaysia is still about 40% under natural forest cover, the main concern is the condition of these remaining forests. As already discussed, after timber harvesting cycles most of these forests may not reflect the original structure and species composition. The need to conserve all forest/vegetation types of the country is crucial if we are to keep intact at least some representation of what nature has taken millions of years to form. Representative conserved areas should not only be limited to just the known forest types in restricted protected areas but should be distributed throughout the country. The current totally protected areas as represented in National and State parks, and Wildlife sanctuaries are not sufficient to represent such distribution. Plant distribution is not just limited by the climatic and edaphic factors as described in this chapter. The geological past also has a tremendous influence on plant distribution. To conserve both the representative forest and vegetation types in the country together with unique geographical differences in the local flora, it makes sense to set up a conservation network throughout the country. For forests, relatively small areas of good forests even if they have been previously logged, can conserve important representation of the former forest types and local species variation. A major conservation aid towards this end will be for all forest reserves in the country to have within the reserves, areas identified for conservation purposes. Such ideas were already implemented under the Virgin Jungle Reserve system (Wyatt-Smith, 1963). Where virgin areas are lacking, good patches of logged-over forests can provide such a function. The initiation of High Conservation Value Forest in recent years under the forest certification scheme can be used for this purpose. It is, however, imperative that identification of such sites be done quickly and steps taken to protect the remaining sites that are suitable for conservation.

References

Anderson, J.A.R. 1963. The flora of the peat swamp forests of Sarawak and Brunei including a catalogue of all recorded species of flowering plants, ferns and fern allies. *Gard. Bull. Sing*. 20: 131–228.

Anderson, J.A.R. 1964. The structure and development of the peat swamp forest in Sarawak and Brunei. *J. Trop. Geog*. 18: 7–16.

Anon. 2008. *Common Vision on Biodiversity - In Government and Development Processes. Reference Document for Planners, Decision-makers and Practioners*. Pp. 1-111. Ministry of Natural Resources and Environment, Kuala Lumpur.

Appanah, S. 2000. Trends and Issues in Tropical Forest Management: Setting the Agenda for Malaysia. Pp 1–21. In *Proceedings of the Conference on Tropical Forest Harvesting*: *New Technologies Examined*, 22-24 Nov. 1999. Conference on Forestry and Forest Products Research 1999 Series. Forest Research Institute Malaysia.

Ashton, P.S. 1995. Biogeography and Ecology. Pp. XLIII–LI. In: Soepadmo, E. & Wong, K.M. (eds.) *Tree Flora. Sabah and Sarawak Volume 1*. Forest Research Institute Malaysia, Kuala Lumpur, Sabah Forestry Department, Sandakan & Sarawak Forestry Department, Kuching.

Ashton, P.S. 2008. Changing values of Malaysian forests: the challenge of biodiversity and its sustainable management. *J. Trop. For. Sci*. 20: 282–291.

Brunig, E.F. 1974. *Ecological Studies in the Kerangas Forests of Sarawak and Brunei*. Borneo Literature Bureau, Kuching.

Chew M.Y. & Siti-Munirah, M.Y. (2010) Ecological implications from the naturalisation of noxious *Cabomba* waterweeds in Malaysia. *Malay. Naturalist.* 63(2): 19–21.

Chin, S.C. 1977. The limestone hill flora of Malaya. I. *Gard. Bull. Sing*. 30: 165–219.

Chin, T.Y., Nor Akhiruddin, M., Samsuanuar, N., Yong, T.K., Hasnuddin, M.A. & Mohd Nashir, S.I. 1997. *Inventori Hutan Nasional Ketiga*, *Semenanjung Malaysia*. Jabatan Perhutanan Semenanjung Malaysia (in Malay).

Chua, L.S.L. & Saw, L.G. 2001. A reassessment of the flora of Gunung Ulu Kali, Genting Highlands, Malaysia – Preliminary findings and trends. *Malay. Nat. J*. 55: 65–76.

Corner, E.J.H. 1978. The freshwater swamp-forest of South Johore and Singapore. *Gard. Bull. Sing.* Suppl. 1: 1–266.

Corner, E.J.H. 1985. The botany of some islets of east of Pahang. *Gard. Bull. Sing*. 38:1–42.

Gregory, R. 2006. *The Rise and Fall of Lake Chini.* Wild Asia. http://www.wildasia.org/main.cfm/support/Lake_Chini (downloaded 15 July 2009).

Henderson, M.R. 1935. The epiphytic flora of *Dipterocarpus oblongifolius* Blume (Neram). *Gard. Bull. S.S.* 9: 93–97.

Hartog, C. den. 1970. The seagrasses of the world. Tweede Reeks, Deel 59 (1). North Holland Publishing Company, Amsterdam.

Japar Sidik, B., Muta Harah, Z. & Aziz, A. 2006. Distribution and significance of seagrass ecosystems in Malaysia. *Aquatic Ecosystem Health & Management* 9: 203–214.

Kiew, R. 1982. The Klang Gates Ridge. *Malay. Nat.* 36(1): 22–28.

Kiew, R. 1998. The Seed Plant Flora of Fraser's Hill. Research Pamphlet No. 121. Forest Research Institute Malaysia, Kuala Lumpur.

Kiew, R. 2005. *Begonias of Peninsular Malaysia*. Natural History Publications (Borneo) in association with Singapore Botanic Gardens, Kota Kinabalu.

Kiew, R. 2009. Additions to the weed flora of Peninsular Malaysia. *Malay. Nat. J*. 61: 133–142.

Lim, R.P., Furtado, J.I. & Morley, R.J. 1982. General description of Tasek Bera. Pp. 7–54. In: J.I. Furtado & S. Mori (eds). *Tasek Bera: the Ecology of a Freshwater Swamp*. Junk Publ., The Hague.

Manokaran, N., Quah, E.S., Ashton, P.S., LaFrankie, J.V., Nur Supardi, M.N., Wan Mohd Sukri, W.A. & Okuda, T. 2004. Pasoh forest dynamics plot, Peninsular Malaysia. Pp. 585–598. In: E.C. Losos & E.G. Leigh Jr. (eds). *Tropical Forest Diversity and Dynamism.* The University of Chicago Press, Chicago and London.

Ng, T.P. & Shamsudin, I. 2001. Common trees in peat swamp forests of Peninsular Malaysia. Research Pamphlet No. 124. Forest Research Institute Malaysia, Kuala Lumpur.

NRE. 2008. *A Common Vision on Biodiversity – In Government and the Development Process.* The Ministry of Natural Resources and Environment, Putrajaya.

Phang, S.M. 2000. *Seagrasses of Malaysia*. University of Malaya Botanical Monographs No. 2. University of Malaya. Kuala Lumpur

Plotkin, J.B., Potts, M.D., Yu, D.W., Bunyavejchewin, S., Condit, R., Foster, R., Hubbell, S., LaFrankie, J., Manokaran, N., Lee, H.S., Sukumar, R., Nowak, M.A. & Ashton, P.S. 2000. Predicting species diversity in tropical forests. *PNAS* 97: 10850–10854.

Primack,, R.B & Lee, H.S. 1991. Population dynamics of pioneer (*Macaranga*) trees an understorey (*Mallotus*) trees (Euphorbiaceae) in primary and secondary logged Bornean forests. *J. Trop. Ecol*. 7: 439–458.

Rafidah, A.R., Chew, M.Y., Ummul-Nazrah, A.R. & Kamarudin, S. 2010. The botany of Tasik Bera, Pahang, Malaysia. *Malay. Nat. J.* 62: 249–267.

Ridley, H.N. 1915. The botany of Gunong Tahan. *J. Fed. Malay St. Mus*. 8: 127–202.

Saw, L.G. 2004. Tropical forests/Tropical moist forests. Pp. 1775–1782. In: J. Burley (ed.) *Encyclopedia of Forest Sciences*. Elsevier, Amsterdam.

Saenger, P., Hegerl, E.J. & Davie, J.D.S. 1983. *Global Status of Mangrove Ecosystems.* Commission on Ecological Papers No. 3. International Union for Conservation of Nature and Natural Resources (IUCN), The Netherlands.

Soepadmo, E. 1971. Plants and vegetation along Sungai Tahan from Kuala Tahan to Gunong Tahan. *Malay. Nat. J*. 24: 118–124.

Steenis, C.G.G.J. van 1981. *Rheophytes of the World*. Sijthoff and Noordhoff, Alpena an den Rijn.

Stone, B.C. 1981. The summit flora of Gunung Ulu Kali (Pahang, Malaysia). *Fed. Mus. J.* 26: 1–157.

Symington, C F. 1943 (Revised by Ashton, P.S. & Appanah, S., 2004). Foresters' Manual of Dipterocarps. *Malay. For. Records* No. 16. University of Malaya Press, Kuala Lumpur.

Tjia, H.D. 1988. The physical setting. Pp. 1–19. In Earl of Cranbrook (ed.). *Key Environments: Malaysia.* Pergamon Press. Oxford.

Wan Mohd Shukri, W.A., Wan Juliana, W.A. & Samsudin, M. 2006. Selective Management System (SMS): Is it a success? Pp. 60–68. In: M.J. Jamaluddin, A. Kadir, A. Azahan & R. Muhammad Rizal (eds). *Proceeding of the 3rd Bangi World Conference on Environmental Management: Managing Changes*, Universiti Kebangsaan Malaysia Press, Bangi.

Wan Mohd Shukri, W.A., Ismail, H., Nur Hajar, Z.S. & Mohd Rizal, S. 2007. Logged-over forests under the selective management system: A case study in Jengai Forest Reserve, Terengganu. Pp. 68–77. In: N.M Nik Zanariah, K.A. Sarifah & H. Nor Azman (eds). *Highlights of FRIM'S IRPA Projects 2006*. Forest Research Institute Malaysia, Kepong.

Wan Mohd Shukri, W.A., Ismail, H., Abd. Rahman, K. & Hur Hajar, Z.S. 2005. Status of Logged-Over Forest in Compartment 53 and 114 Jengai F.R., Terengganu. Pp. 87–94. In: H. Nor Azman, N.M. Nik Zanariah & H.T. Chan (eds). *Prosiding Seminar Hasil Kajian IRPA RMK-8 2004 FRIM*. Forest Research Institute Malaysia, Kepong.

Watson, J.G. 1928. Mangrove Forests of the Malay Peninsula. *Malayan Forest Records No. 6*. Federated Malay States Government, Singapore.

Webb, L.J., Tracey, J.G., Williams, W.T. & Lance, G.N. 1967. Studies in the numerical analysis of complex rain forest communities. I. A comparison of methods applicable to site/species data. *J. Ecol.* 55: 171–191.

Whitmore, T.C. 1984. *Tropical Rain Forests of the Far East.* Second Edition. ELBS/Oxford University Press, Oxford.

Wong, K.M. 1995. The Bamboos of Peninsular Malaysia. *Malayan Forest Records No. 41*. Forest Research Institute Malaysia, Kuala Lumpur.

Wong, K.M., Saw, L.G. & Kochummen, K.M. 1987. A survey of the forests of the Endau-Rompin area, Peninsular Malaysia: Principal forest types and floristic notes. *Malay. Nat. J.* 41: 125–144.

Wyatt-Smith, J. 1953. A note on the vegetation of some islands in the Malacca Straits. *Malay. For.* 16:191–205.

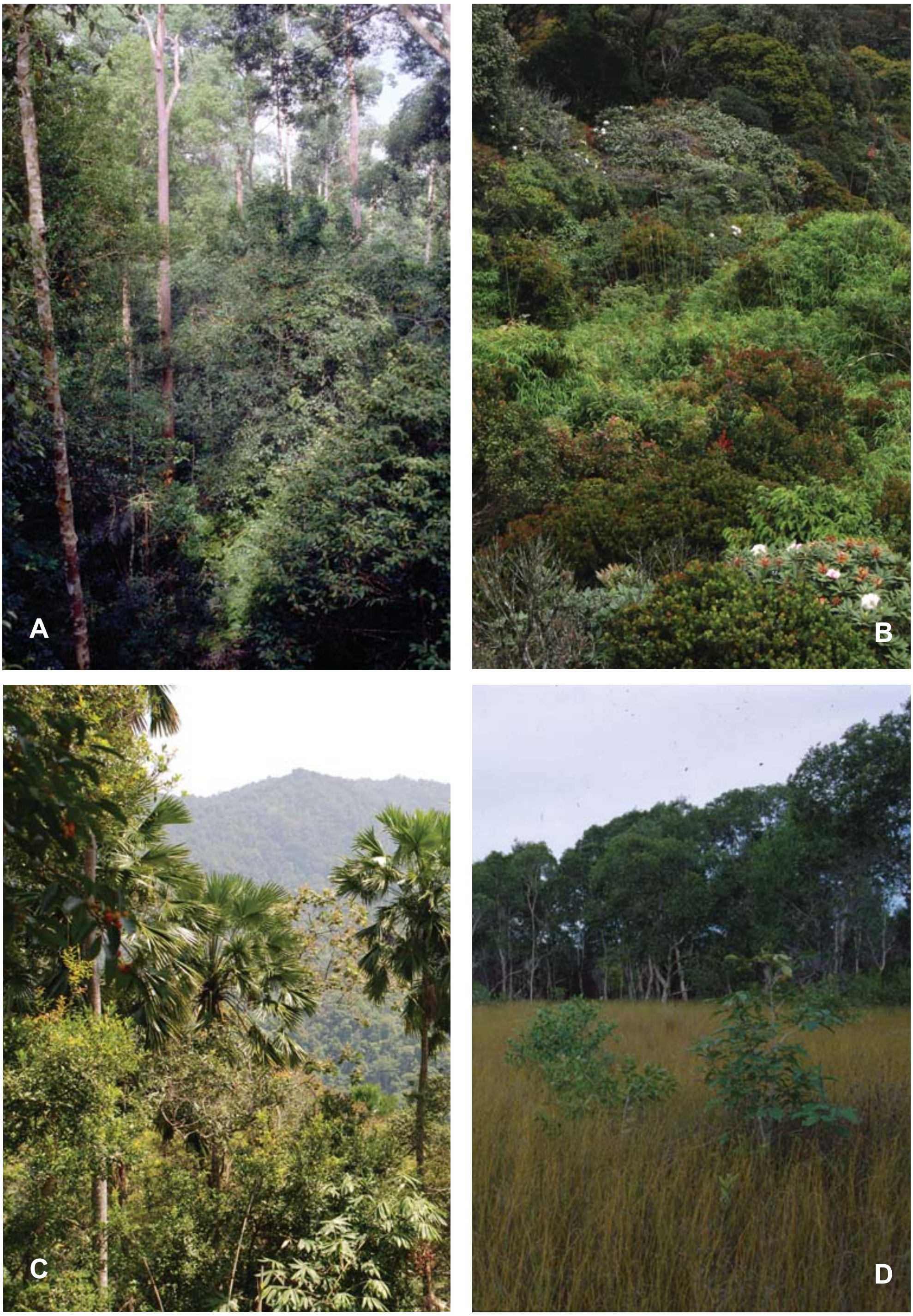
Wyatt-Smith, J. 1963. Manual of Malayan silviculture for inland forest. *Malayan Forest Records No. 23*. Vol. 1. Forest Research Institute, Kuala Lumpur.

Plate 1. A, lowland dipterocarp forest, Endau-Rompin State Park. B, hill dipterocarp forest, Maxwell Hill, the glaucous crowns on the ridges are *Shorea curtisii.*

L.G. Saw

L.G. Saw

Plate 2. A, semi-evergreen forest, Machinchang FR, Langkawi. B, lower montane forest, Fraser's Hill.



L.G. Saw
M.Y. Chew
L.G. Saw
L.G. Saw

Plate 3. A, lowland dipterocarp forest, Pasoh FR, note the emergents and middle canopy layer. B, upper montane forest, G. Brinchang, Cameron Highlands. *Rhododendron wrayi* with white blooms (foreground) and thickets of bamboo, *Holttumochloa magica.* C, stands of *Livistona speciosa* on ridges of upper hill dipterocarp forest. D, heath vegetation on raised sandbanks, Jambu Bongkok, Terengganu.

L.G. Saw

M.Y Chew

L.G. Saw

L.G. Saw

Plate 4. A, heath vegetation on sandstone, Endau-Rompin State Park. B, heath vegetation, G. Tahan. C, *Livistona endauensis* stand on sandstone massif, Endau Rompin State Park. D, limestone hill, Batu Caves.

Plate 5. A, beach vegetation, Langkawi. B, freshwater swamp forest, rassau (*Pandanus helicopus*)-belt.

Plate 6. A, mangrove forest, stand of *Rhizophora apiculata*, Matang FR. B, mangrove forest, stand of *Bruguiera gymnorhiza*, Tioman Island. C, brackish water vegetation, *Nypa fruticans* in the foreground, *Phoenix paludosa* in the background, Marang, Terengganu. D, freshwater swamp forest, *Pholidocarpus kingianus*, Seri Iskandar, Perak.

Plate 7. A, riparian vegetation, rheophytes on stream banks. B, aquatic vegetation in natural lake, Tasik Chini.

Plate 8. A, *Syzygium* swamp in Tasik Bera. B, quartz ridge, Klang Gates. C, *Baeckea frutescens* on quartz ridge, Klang Gates.